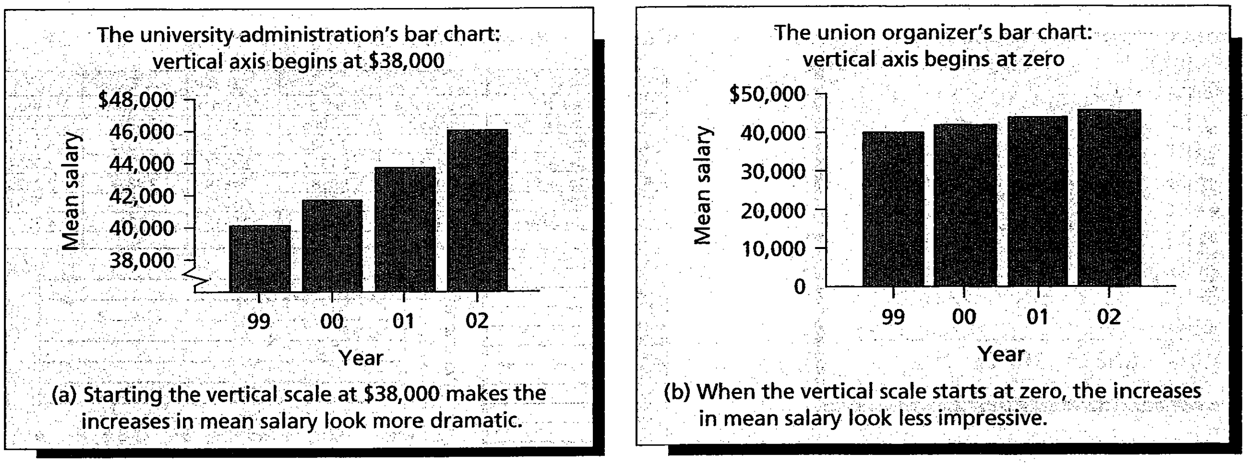
Bruce L. Bowerman, Richard T. O'Connell,
Business Statistics in Practice, Third Edition, IE,
McGraw-Hill, Irwin, 2003.
pp. 93-97

*2.7 Misleading Graphs and Charts

The statistical analyst's goal should be to present the most accurate and truthful portrayal of a data set that is possible. Such a presentation allows managers using the analysis to make informed decisions. However, it is possible to construct statistical summaries that are misleading. Although we do not advocate using misleading statistics, you should be aware of some of the ways statistical graphs and charts can be manipulated in order to distort the truth. By knowing

FIGURE 2.50 **Two Bar Charts of the Mean Salaries at a Major University from 1999 to 2002**



(a) Starting the vertical scale at $38,000 makes the increases in mean salary look more dramatic.

(b) When the vertical scale starts at zero, the increases in mean salary look less impressive.

what to look for, you can avoid being misled by a (we hope) small number of unscrupulous practitioners.

As an example, suppose that the faculty at a major university will soon vote on a proposal to join a union. Both the union organizers and the university administration plan to distribute recent salary statistics to the entire faculty. Suppose that the mean faculty salary at the university and the mean salary increase at the university (expressed as a percentage) for each of the years 1999 through 2002 are as follows:

Year	Mean Salary (All Ranks)	Mean Salary Increase (Percent)
1999	$40,000	3.0%
2000	41,600	4.0
2001	43,500	4.5
2002	46,100	6.0

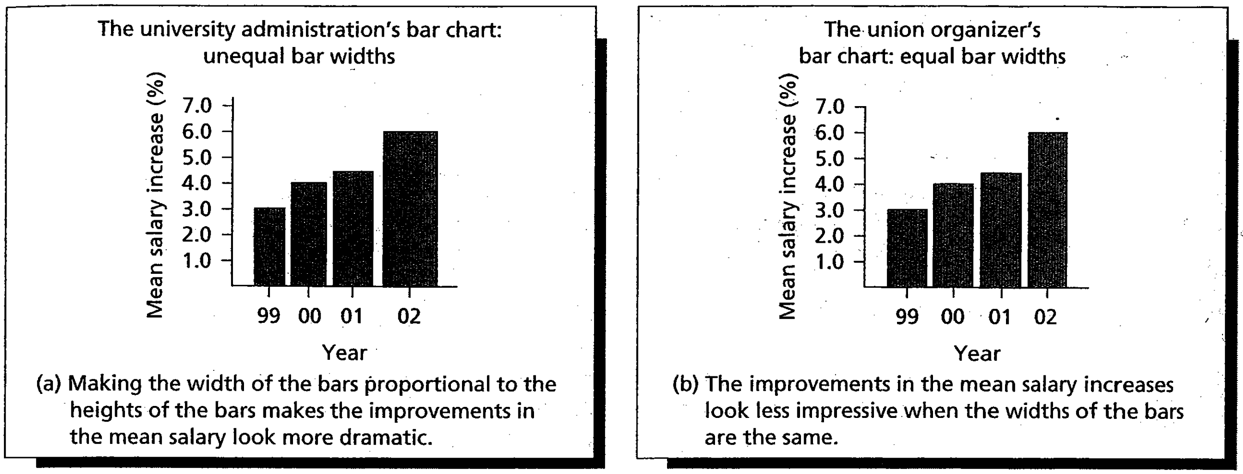
The university administration does not want the faculty to unionize and, therefore, hopes to convince the faculty that substantial progress has been made to increase salaries without a union. On the other hand, the union organizers wish to portray the salary increases as minimal so that the faculty will feel the need to unionize.

Figure 2.50 gives two bar charts of the mean salaries at the university for each year from 1999 to 2002. Notice that in Figure 2.50(a) the administration has started the vertical scale of the bar chart at a salary of $38,000 by using a *scale break* (⌇). Alternatively, the chart could be set up without the scale break by simply starting the vertical scale at $38,000. Starting the vertical scale at a value far above zero makes the salary increases look more dramatic. Notice that when the union organizers present the bar chart in Figure 2.50(b), which has a vertical scale starting at zero, the salary increases look far less impressive.

Figure 2.51 presents two bar charts of the mean salary increases (in percentages) at the university for each year from 1999 to 2002. In Figure 2.51(a), the administration has made the widths of the bars representing the percentage increases proportional to their heights. This makes the upward movement in the mean salary increases look more dramatic because the observer's eye tends to compare the areas of the bars, while the improvements in the mean salary increases are really only proportional to the heights of the bars. When the union organizers present the bar chart of Figure 2.51(b), the improvements in the mean salary increases look less impressive because each bar has the same width.

Figure 2.52 gives two runs plots (also called **time series plots**) of the mean salary increases at the university from 1999 to 2002. In Figure 2.52(a) the administration has stretched the vertical axis of the graph. That is, the vertical axis is set up so that the distances between the percentages are large. This makes the upward trend of the mean salary increases appear to be steep. In Figure 2.52(b) the union organizers have compressed the vertical axis (that is, the distances between

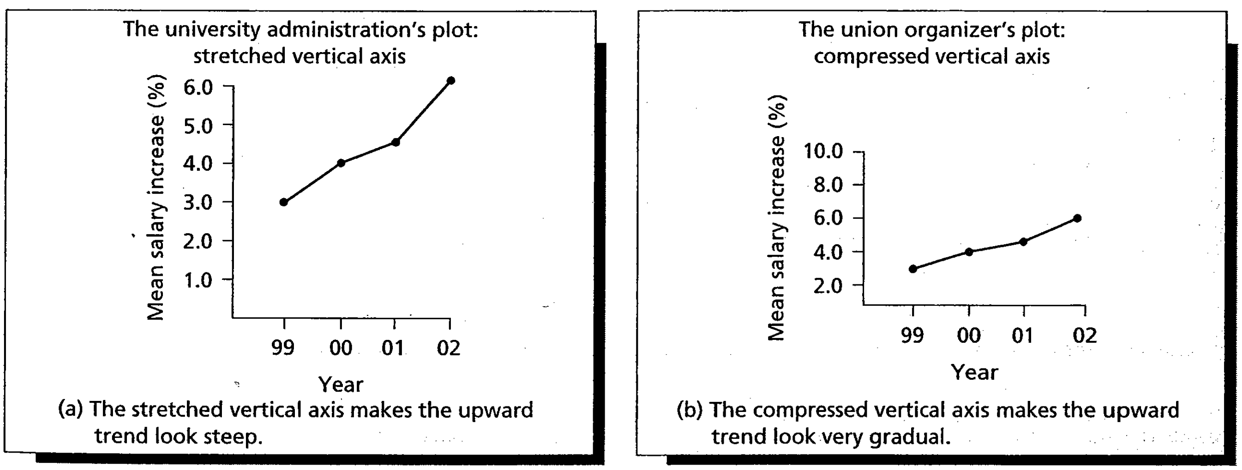
FIGURE 2.51 Two Bar Charts of the Mean Salary Increases at a Major University from 1999 to 2002



(a) Making the width of the bars proportional to the heights of the bars makes the improvements in the mean salary look more dramatic.

(b) The improvements in the mean salary increases look less impressive when the widths of the bars are the same.

FIGURE 2.52 Two Graphs of the Mean Salary Increases at a Major University from 1999 to 2002



(a) The stretched vertical axis makes the upward trend look steep.

(b) The compressed vertical axis makes the upward trend look very gradual.

the percentages are small). This makes the upward trend of the mean salary increases appear to be gradual. As we will see in the exercises, stretching and compressing the horizontal axis in a runs plot can also greatly affect the impression given by the plot.

It is also possible to create totally different interpretations of the same statistical summary by simply using different labeling or captions. For example, consider the bar chart of mean salary increases in Figure 2.51(b). To create a favorable interpretation, the university administration might use the caption "Salary Increase Is Higher for the Fourth Year in a Row." On the other hand, the union organizers might create a negative impression by using the caption "Salary Increase Fails to Reach 10% for Fourth Straight Year."

In summary, we do not approve of using statistics to mislead and distort reality. Statistics should be used to present the most truthful and informative summary of the data that is possible. However, it is important to carefully study any statistical summary so that you will not be misled. Look for manipulations such as stretched or compressed axes on graphs, axes that do not begin at zero, and bar charts with bars of varying widths. Also, carefully think about assumptions, and make your own conclusions about the meaning of any statistical summary rather than relying on captions written by others. Doing these things will help you to see the truth and to make well-informed decisions.

Exercises for Section 2.7

CONCEPTS

2.64 When we construct a bar chart or graph, what is the effect of starting the vertical axis at a value that is far above zero? Explain.

2.65 Find an example of a misleading use of statistics in a newspaper, magazine, corporate annual report, or other source. Then explain why your example is misleading.

METHODS AND APPLICATIONS

2.66 Figure 2.53 gives two more time series plots of the previously discussed salary increases. In Figure 2.53(a) the administration has compressed the horizontal axis. In Figure 2.53(b) the union organizers have stretched the horizontal axis. Discuss the different impressions given by the two time series plots.

2.67 In the article "How to Display Data Badly" in the May 1984 issue of *The American Statistician,* Howard Wainer presents a *stacked bar chart* of the number of public and private elementary schools (1929–1970). This bar chart is given in Figure 2.54. Wainer also gives a line graph of the number of private elementary schools (1930–1970). This graph is shown in Figure 2.55.

FIGURE 2.53 Two Graphs of the Mean Salary Increases at a Major University from 1999 to 2002

(a) The administration's plot: compressed horizontal axis **(b) The union organizers' plot: stretched horizontal axis**

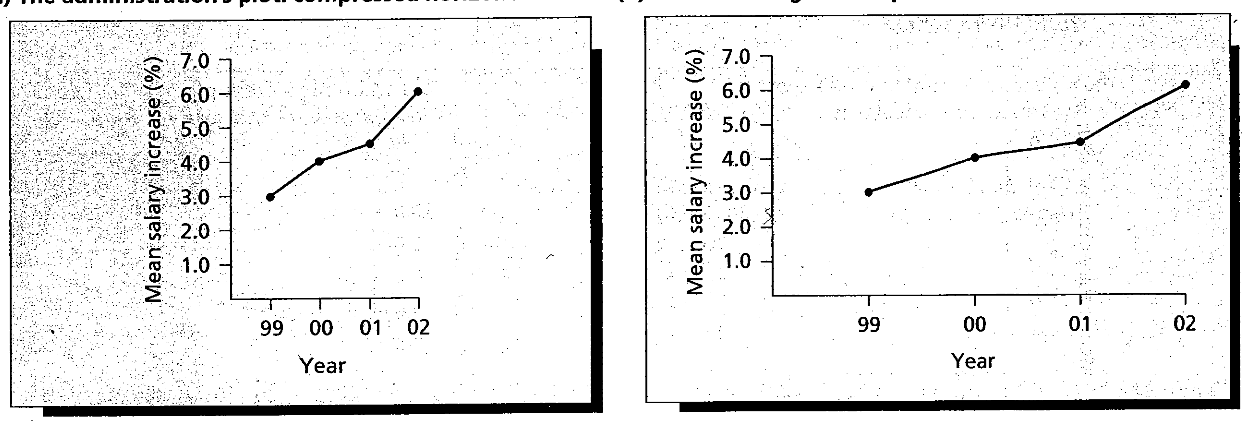


FIGURE 2.54 Wainer's Stacked Bar Chart

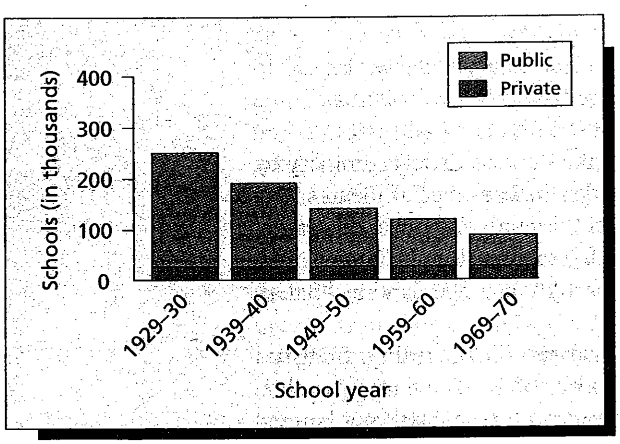
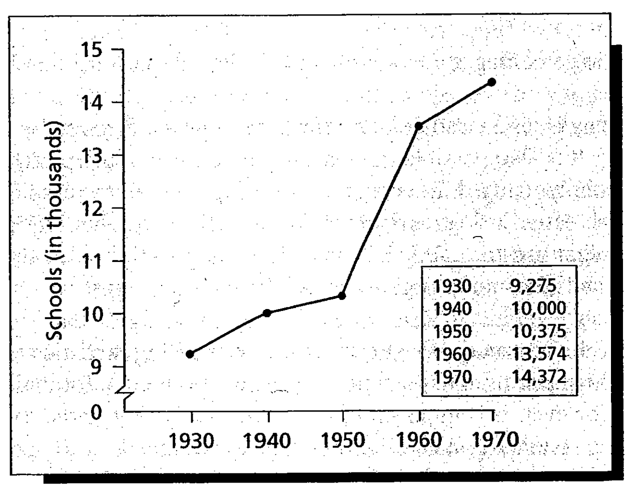


FIGURE 2.55 Wainer's Line Graph



a Looking at the bar chart of Figure 2.54, does there appear to be an increasing trend in the number of private elementary schools from 1930 to 1970?
b Looking at the line graph of Figure 2.55, does there appear to be an increasing trend in the number of private elementary schools from 1930 to 1970?
c Which portrayal of the data do you think is more appropriate? Explain why.
d Is either portrayal of the data entirely appropriate? Explain.